



Implementación de VPN WireGuard entre Servidores en Google Cloud Platform

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▼ Descripción General

Esta guía detalla la implementación de una conexión VPN segura usando WireGuard entre dos servidores ubicados en diferentes cuentas de Google Cloud Platform. WireGuard es una VPN moderna, rápida y segura que utiliza criptografía de última generación.

Objetivos

- Configurar dos instancias de VM en cuentas GCP separadas
- Implementar WireGuard como solución VPN

- Establecer un túnel seguro de comunicación entre los servidores
- Verificar la conectividad y transferencia segura de datos

▼ Arquitectura de la Solución

```
graph TB
    subgraph "Cuenta GCP Principal/Servidor"
        A["VM Servidor WireGuard<br/>10.0.1.2/24<br/>WG: 10.100.0.1/24"]
        B["VPC Network<br/>10.0.1.0/24"]
        C["Firewall Rules<br/>UDP:51820"]
        A --> B
        B --> C
    end

    subgraph "Cuenta GCP Tercero/Cliente"
        D["VM Cliente WireGuard<br/>10.0.2.2/24<br/>WG: 10.100.0.2/24"]
        E["VPC Network<br/>10.0.2.0/24"]
        F["Firewall Rules<br/>UDP:51820"]
        D --> E
        E --> F
    end

    C <--> |"Túnel WireGuard<br/>Puerto UDP 51820<br/>Encriptado"| F

    style A fill:#D6EAF8,stroke:#2980B9,color:#1B2631
    style D fill:#D5F5E3,stroke:#27AE60,color:#1B2631
    style B fill:#FDFEFE,stroke:#BDC3C7,color:#2C3E50
    style E fill:#FDFEFE,stroke:#BDC3C7,color:#2C3E50
    style C fill:#FCF3CF,stroke:#F1C40F,color:#7D6608
    style F fill:#FCF3CF,stroke:#F1C40F,color:#7D6608
```

▼ Componentes Principales

1. Servidor WireGuard (Cuenta Principal)

- IP Pública: Asignada por GCP
- IP Privada VPC: 10.0.1.2/24
- IP WireGuard: 10.100.0.1/24
- Puerto: UDP 51820

2. Cliente WireGuard (Cuenta Tercero)

- IP Pública: Asignada por GCP
- IP Privada VPC: 10.0.2.2/24
- IP WireGuard: 10.100.0.2/24

▼ Prerrequisitos

- Dos cuentas de Google Cloud Platform (Free Tier)
- Conocimientos básicos de Linux y redes
- Acceso a Google Cloud Console o gcloud CLI

▼ Paso 1: Configuración del Entorno

1.1 Configuración de la Cuenta Principal

```
# Crear el proyecto
gcloud projects create mi-proyecto-vpn-460916 --name="Mi Proyecto VPN"
gcloud config set project mi-proyecto-vpn-460916

# Habilitar servicios de computo
gcloud services enable compute.googleapis.com

# Crear VPC
gcloud compute networks create vpc-propia \
  --subnet-mode=custom \
  --bgp-routing-mode=regional

# Crear subnet
gcloud compute networks subnets create subnet-propia \
  --network=vpc-propia \
  --region=us-central1 \
  --range=10.0.1.0/24

# Crear regla firewall para WireGuard - ssh
gcloud compute firewall-rules create allow-wireguard \
  --network=vpc-propia \
  --allow=udp:51820 \
```

```

--source-ranges=0.0.0.0/0

gcloud compute firewall-rules create allow-ssh \\\
  --network=vpc-propia \\\
  --allow=tcp:22 \\\
  --source-ranges=0.0.0.0/0

# Crear instancia principal
gcloud compute instances create wg-server \\\
  --zone=us-central1-a \\\
  --machine-type=e2-micro \\\
  --network-interface=network=vpc-propia,subnet=subnet-propia \\\
  --image-family=ubuntu-2004-lts \\\
  --image-project=ubuntu-os-cloud \\\
  --boot-disk-size=10GB \\\
  --tags=wireguard-server

```

1.2 Configuración de la Cuenta Tercero

Repetimos los mismos pasos pero con diferentes nombres y rangos de IP (para diferenciarlos a simple vista):

```

# Crear el proyecto
gcloud projects create tercero-proyecto-vpn --name="Tercero Proyecto VPN"
gcloud config set project tercero-proyecto-vpn

# Habilitar servicios de computo
gcloud services enable compute.googleapis.com

# Crear VPC
gcloud compute networks create vpc-tercero \\\
  --subnet-mode=custom \\\
  --bgp-routing-mode=regional

# Crear subnet
gcloud compute networks subnets create subnet-tercero \\\
  --network=vpc-tercero \\\

```

```

--region=us-central1 \\\n
--range=10.0.2.0/24\n\n
# Crear reglas de firewall\n
gcloud compute firewall-rules create allow-wireguard-tercero \\\n
--network=vpc-tercero \\\n
--allow=udp:51820 \\\n
--source-ranges=0.0.0.0/0\n\n
gcloud compute firewall-rules create allow-ssh-tercero \\\n
--network=vpc-tercero \\\n
--allow=tcp:22 \\\n
--source-ranges=0.0.0.0/0\n\n
# Crear instancia del Cliente\n
gcloud compute instances create wg-client \\\n
--zone=us-central1-a \\\n
--machine-type=e2-micro \\\n
--network-interface=network=vpc-tercero,subnet=subnet-tercero \\\n
--image-family=ubuntu-2004-lts \\\n
--image-project=ubuntu-os-cloud \\\n
--boot-disk-size=10GB \\\n
--tags=wireguard-client

```

▼ Paso 2: Generación de Certificados y Claves

2.1 En el Servidor (Cuenta Propia)

```

# Conectarse al servidor\n
gcloud compute ssh wg-server --zone=us-central1-a\n\n
# Instalar WireGuard\n
sudo apt update\n
sudo apt install -y wireguard\n\n
# Generar claves del servidor\n
cd /etc/wireguard\n
umask 077

```

```
wg genkey | sudo tee server_private.key
sudo cat server_private.key | wg pubkey | sudo tee server_public.key

# Guardar las claves
SERVER_PRIVATE_KEY=$(sudo cat server_private.key)
SERVER_PUBLIC_KEY=$(sudo cat server_public.key)

echo "Clave privada del servidor: $SERVER_PRIVATE_KEY"
echo "Clave pública del servidor: $SERVER_PUBLIC_KEY"
```

2.2 En el Cliente (Cuenta Tercero)

```
# Conectarse al cliente
gcloud compute ssh wg-client --zone=us-central1-a

# Instalar WireGuard
sudo apt update
sudo apt install -y wireguard

# Generar claves del cliente
cd /etc/wireguard
umask 077
wg genkey | sudo tee client_private.key
sudo cat client_private.key | wg pubkey | sudo tee client_public.key

# Guardar las claves
CLIENT_PRIVATE_KEY=$(sudo cat client_private.key)
CLIENT_PUBLIC_KEY=$(sudo cat client_public.key)

echo "Clave privada del cliente: $CLIENT_PRIVATE_KEY"
echo "Clave pública del cliente: $CLIENT_PUBLIC_KEY"
```

▼ Paso 3: Configuración del Servidor VPN (Principal)

3.1 Creamos un archivo de configuración del servidor

En el servidor principal, creamos el archivo `/etc/wireguard/wg0.conf` :

```
sudo nano /etc/wireguard/wg0.conf
```

Contenido del archivo(reemplazar los valores entre corchetes)::

```
[Interface]
# Dirección IP del servidor en la VPN
Address = 10.100.0.1/24
# Puerto de escucha
ListenPort = 51820
# Clave privada del servidor
PrivateKey = [SERVER_PRIVATE_KEY]
# Comandos post-up y post-down para NAT (opcional)
PostUp = iptables -A FORWARD -i wg0 -j ACCEPT; iptables -t nat -A POSTROUTING -o ens4 -j MASQUERADE
PostDown = iptables -D FORWARD -i wg0 -j ACCEPT; iptables -t nat -D POSTROUTING -o ens4 -j MASQUERADE

[Peer]
# Cliente
PublicKey = [CLIENT_PUBLIC_KEY]
# IP permitida del cliente
AllowedIPs = 10.100.0.2/32
```

Remplazando las variables quedaria algo asi:

```
[Interface]
Address = 10.100.0.1/24
ListenPort = 51820
PrivateKey = SHaxS5zYYiqJJzeKu+yatyeZBTrgCwBg/Q57yt6pdU4=
PostUp = iptables -A FORWARD -i wg0 -j ACCEPT; iptables -t nat -A POSTROUTING -o ens4 -j MASQUERADE
PostDown = iptables -D FORWARD -i wg0 -j ACCEPT; iptables -t nat -D POSTROUTING -o ens4 -j MASQUERADE

[Peer]
PublicKey = +aJT4Fj/b27UIZXoBuAwEoli2i9QNo0cySYjG3+Oywe=
AllowedIPs = 10.100.0.2/32
```

3.2 Habilitamos el forwarding de IP

```
# Habilitar IP forwarding
sudo sysctl -w net.ipv4.ip_forward=1
echo "net.ipv4.ip_forward=1" | sudo tee -a /etc/sysctl.conf

# Iniciar WireGuard
sudo systemctl enable wg-quick@wg0
sudo systemctl start wg-quick@wg0

# Verificar estado
sudo wg show
```

▼ Paso 4: Configuración del Cliente VPN

4.1 Obtener IP pública del servidor WireGuard (Cuenta Principal)

En el servidor WireGuard (wg-server en la cuenta principal):

```
# Seleccionamos el proyecto y nos conectamos ssh a la instancia
gcloud config set project mi-proyecto-vpn-460916
gcloud compute ssh wg-server --zone=us-central1-a

# Obtenemos la ip publica
curl -s ifconfig.me

# Alternativa desde Cloud Shell sin conectarse a la instancia:
gcloud compute instances describe wg-server \\\
  --zone=us-central1-a \\\
  --project=mi-proyecto-vpn-460916 \\\
  --format='get(networkInterfaces[0].accessConfigs[0].natIP)'

# Guardar IP, por ejemplo: 34.123.45.67
```

4.2 Crear archivo de configuración del cliente en la VM del tercero

En el cliente WireGuard (wg-client en la cuenta del tercero):


```
# Seleccionamos el proyecto y nos conectamos ssh a la instancia
gcloud config set project tercero-proyecto-vpn
gcloud compute ssh wg-client --zone=us-central1-a

# Creamos el archivo de configuración
sudo nano /etc/wireguard/wg0.conf
```

Contenido del archivo (reemplazar los valores entre corchetes):

```
[Interface]
# Dirección IP del cliente en la red VPN
Address = 10.100.0.2/24
# Clave privada del cliente (generada en el Paso 2.2)
PrivateKey = [CLAVE_PRIVADA_DEL_CLIENTE]
# DNS (opcional)
DNS = 8.8.8.8

[Peer]
# Información del servidor (cuenta principal)
# Clave pública del servidor (generada en el Paso 2.1)
PublicKey = [CLAVE_PUBLICA_DEL_SERVIDOR]
# IP pública del servidor:Puerto (obtenida en 4.1)
Endpoint = [IP_PUBLICA_DEL_SERVIDOR]:51820
# Rango de IPs que se enrutarán a través del túnel
AllowedIPs = 10.100.0.0/24
# Mantener la conexión activa (envía paquete cada 25 segundos)
PersistentKeepalive = 25
```

Remplazando las variables quedaria algo asi:

```
[Interface]
Address = 10.100.0.2/24
PrivateKey = yGkpeuF2MYJRF3lmgR01dyyfe9hs/2X0EuC3d1LuP3w=
DNS = 8.8.8.8

[Peer]
PublicKey = OngJgxrlBXdmWLmk2ZujwwBI0TruYemegsveoB6VDk0=
Endpoint = 34.68.174.176:51820
```

```
AllowedIPs = 10.100.0.0/24
PersistentKeepalive = 25
```

4.3 Iniciar cliente WireGuard en la instancia del tercero

Continuar en el cliente (wg-client):

```
# Asignar los permisos correctos al archivo de configuracion
sudo chmod 600 /etc/wireguard/wg0.conf

# Habilitar el servicio WireGuard
sudo systemctl enable wg-quick@wg0

# Iniciar WireGuard
sudo systemctl start wg-quick@wg0

# Verificar estado del servicio
sudo systemctl status wg-quick@wg0

# Ver información de WireGuard
sudo wg show
```

▼ Paso 5: Establecimiento del Túnel

5.1 Verificar conexión

En el cliente:

```
# Ping al servidor VPN
ping -c 4 10.100.0.1

# Verificar interfaz WireGuard
ip addr show wg0

# Ver estadísticas de WireGuard
sudo wg show
```

En el servidor:

```
# Verificar peers conectados
sudo wg show

# Veamos los logs
sudo journalctl -u wg-quick@wg0 -f
```

5.2 Verificar túnel establecido

```
# En el servidor, verificar tráfico
sudo tcpdump -i wg0 -n

# En el cliente, generar tráfico
ping 10.100.0.1
```

▼ Paso 6: Pruebas y Verificación

6.1 Prueba de conectividad básica

```
# Desde el cliente
ping -c 10 10.100.0.1

# Desde el servidor
ping -c 10 10.100.0.2
```

6.2 Prueba de transferencia de archivos

```
# En el servidor, crear archivo de prueba
echo "Prueba de transferencia VPN" > /tmp/prueba.txt

# Iniciar servidor HTTP simple
cd /tmp && python3 -m http.server 8000

# En el cliente, descargar archivo
wget <http://10.100.0.1:8000/prueba.txt> -O /tmp/prueba_recibida.txt
cat /tmp/prueba_recibida.txt
```

6.3 Prueba de rendimiento

```
# Instalar iperf3 en ambos servidores
sudo apt install -y iperf3

# En el servidor
iperf3 -s

# En el cliente
iperf3 -c 10.100.0.1 -t 30
```

Comandos de diagnóstico

```
# Ver logs detallados
sudo journalctl -u wg-quick@wg0 -n 50

# Verificar rutas
ip route show

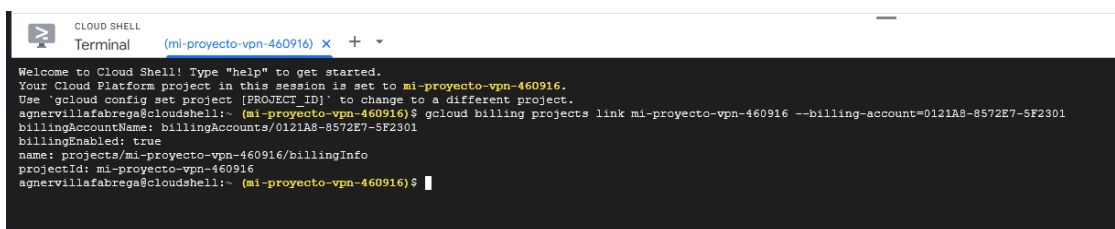
# Verificar interfaces
ip addr show

# Estadísticas de WireGuard
sudo wg show wg0 dump
```

▼ Anexos

▼ Cuenta Principal

Vinculamos la facturación al proyecto (Procedimiento para ambas cuentas)



```
CLOUD SHELL
Terminal (mi-proyecto-vpn-460916) X +
Welcome to Cloud Shell! Type "help" to get started.
Your Cloud Platform project in this session is set to mi-proyecto-vpn-460916.
Use 'gcloud config set project [PROJECT ID]' to change to a different project.
agnervillafabrega@cloudshell:~ (mi-proyecto-vpn-460916)$ gcloud billing projects link mi-proyecto-vpn-460916 --billing-account=0121A8-8572E7-5F2301
billingAccountName: billingAccounts/0121A8-8572E7-5F2301
billingEnabled: true
name: projects/mi-proyecto-vpn-460916/billingInfo
projectId: mi-proyecto-vpn-460916
agnervillafabrega@cloudshell:~ (mi-proyecto-vpn-460916)$
```

Activamos el servicio de computo

```
agnervillafabrega@cloudshell:~ (mi-proyecto-vpn-460916)$ gcloud services enable compute.googleapis.com
Operation "operations/acf.p2-675773673304-719435ab-7e53-44eb-a019-4c156f5bcf06" finished successfully.
agnervillafabrega@cloudshell:~ (mi-proyecto-vpn-460916)$
```

VPC

```
agnervillafabrega@cloudshell:~ (mi-proyecto-vpn-460916)$ gcloud compute networks create vpc-propia \
--subnet-mode=custom \
--bgp-routing-mode=regional
Created [https://www.googleapis.com/compute/v1/projects/mi-proyecto-vpn-460916/global/networks/vpc-propia].
NAME: vpc-propia
SUBNET MODE: CUSTOM
BGP ROUTING MODE: REGIONAL
IPV4 RANGE:
GATEWAY IPV4:
INTERNAL_IPV6_RANGE:

Instances on this network will not be reachable until firewall rules
are created. As an example, you can allow all internal traffic between
instances as well as SSH, RDP, and ICMP by running:

$ gcloud compute firewall-rules create <FIREWALL_NAME> --network vpc-propia --allow tcp,udp,icmp --source-ranges <IP
RANGE>
$ gcloud compute firewall-rules create <FIREWALL_NAME> --network vpc-propia --allow tcp:22,tcp:3389,icmp
```

SUBRED

```
agnervillafabrega@cloudshell:~ (mi-proyecto-vpn-460916)$ gcloud compute networks subnets create subnet-propia \
--network=vpc-propia \
--region=us-central1 \
--range=10.0.1.0/24
Created [https://www.googleapis.com/compute/v1/projects/mi-proyecto-vpn-460916/regions/us-central1/subnetworks/subne
t-propia].
NAME: subnet-propia
REGION: us-central1
NETWORK: vpc-propia
RANGE: 10.0.1.0/24
STACK TYPE: IPV4 ONLY
IPV6 ACCESS TYPE:
INTERNAL_IPV6_PREFIX:
EXTERNAL_IPV6_PREFIX:
agnervillafabrega@cloudshell:~ (mi-proyecto-vpn-460916)$
```

FIREWALL

```
agnervillafabrega@cloudshell:~ (mi-proyecto-vpn-460916)$ gcloud compute firewall-rules create allow-wireguard \
--network=vpc-propia \
--allow=udp:51820 \
--source-ranges=0.0.0.0/0
Creating firewall...working..Created [https://www.googleapis.com/compute/v1/projects/mi-proyecto-vpn-460916/global/f
irewalls/allow-wireguard].
Creating firewall...done.
NAME: allow-wireguard
NETWORK: vpc-propia
DIRECTION: INGRESS
PRIORITY: 1000
ALLOW: udp:51820
DENY:
DISABLED: False
agnervillafabrega@cloudshell:~ (mi-proyecto-vpn-460916)$
```

```
agnervillafabrega@cloudshell:~ (mi-proyecto-vpn-460916)$ gcloud compute firewall-rules create allow-ssh \
--network=vpc-propia \
--allow=tcp:22 \
--source-ranges=0.0.0.0/0
Creating firewall...working..Created [https://www.googleapis.com/compute/v1/projects/mi-proyecto-vpn-460916/global/f
irewalls/allow-ssh].
Creating firewall...done.
NAME: allow-ssh
NETWORK: vpc-propia
DIRECTION: INGRESS
PRIORITY: 1000
ALLOW: tcp:22
DENY:
DISABLED: False
agnervillafabrega@cloudshell:~ (mi-proyecto-vpn-460916)$
```

Creacion de la instancia:

```

agnervillafabrega@cloudshell:~ (mi-proyecto-vpn-460916)$ gcloud compute instances create wg-server \
  --zone=us-central1-a \
  --machine-type=e2-micro \
  --network-interface=network=vpc-propia,subnet=subnet-propia \
  --image-family=ubuntu-2004-lts \
  --image-project=ubuntu-os-cloud \
  --boot-disk-size=10GB \
  --tags=wireguard-server
WARNING: You have selected a disk size of under [200GB]. This may result in poor I/O performance. For more informati
on, see: https://developers.google.com/compute/docs/disks#performance.
Created [https://www.googleapis.com/compute/v1/projects/mi-proyecto-vpn-460916/zones/us-central1-a/instances/wg-serv
er].
NAME: wg-server
ZONE: us-central1-a
MACHINE_TYPE: e2-micro
PREEMPTIBLE:
INTERNAL_IP: 10.0.1.2
EXTERNAL_IP: 34.68.174.176
STATUS: RUNNING
agnervillafabrega@cloudshell:~ (mi-proyecto-vpn-460916)$

```

Conexion e instalacion de wireGuard

Nos conectamos a la instancia principal:

```

agnervillafabrega@cloudshell:~ (mi-proyecto-vpn-460916)$ gcloud compute ssh wg-server --zone=us-central1-a
WARNING: The private SSH key file for gcloud does not exist.
WARNING: The public SSH key file for gcloud does not exist.
WARNING: You do not have an SSH key for gcloud.
WARNING: SSH keygen will be executed to generate a key.
This tool needs to create the directory [/home/agnervillafabrega/.ssh] before being able to generate SSH keys.

Do you want to continue (Y/n)? Y

Generating public/private rsa key pair.
Enter passphrase (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /home/agnervillafabrega/.ssh/google_compute_engine
Your public key has been saved in /home/agnervillafabrega/.ssh/google_compute_engine.pub
The key fingerprint is:
SHA256:zd30jQ/bT9EBPPq5s9DC3mofEIImP97QGNmz6wq5oIc agnervillafabrega@cs-352718124107-default
The key's randomart image is:
+---[RSA 3072]-----+
|      ..      |
|      + . o.  |
|     . = + . . o. |
|    + +o.o.o o+ |
|   =Soo.o =.+ |
|  oE+ o oo =. |
| o.. o + oo + |
| oo . .+o.. |
| .o.....oo+o . |
+---[SHA256]-----+
Updating project ssh metadata...working.Updated [https://www.googleapis.com/compute/v1/projects/mi-proyecto-vpn-460916]
Updating project ssh metadata...done.
Waiting for SSH key to propagate.
Warning: Permanently added 'compute.2726188644038539560' (ED25519) to the list of known hosts.
Welcome to Ubuntu 20.04.6 LTS (GNU/Linux 5.15.0-1081-gcp x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Sun May 25 16:53:12 UTC 2025

System load:  0.0          Processes:           103
Usage of /:   21.4% of 9.51GB Users logged in:       0
Memory usage: 21%         IPv4 address for ens4: 10.0.1.2
Swap usage:   0%

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

```

Instalacionde wireGuard:

```

agnervillafabrega@wg-server:~$ sudo apt install -y wireguard
Reading package lists... Done
Building dependency tree
Reading state information... Done
The following additional packages will be installed:
  wireguard-tools
Suggested packages:
  openresolv | resolvconf
The following NEW packages will be installed:
  wireguard wireguard-tools
0 upgraded, 2 newly installed, 0 to remove and 7 not upgraded.
Need to get 86.6 kB of archives.
After this operation, 344 kB of additional disk space will be used.
Get:1 http://us-central1-gce.archive.ubuntu.com/ubuntu focal-updates/universe amd64 wireguard-tools amd64 1.0.20200513-1-20.04.2 [83.3 kB]
Get:2 http://us-central1-gce.archive.ubuntu.com/ubuntu focal-updates/universe amd64 wireguard all 1.0.20200513-1-20.04.2 [3264 B]
Fetched 86.6 kB in 0s (703 kB/s)
Selecting previously unselected package wireguard-tools.
(Reading database ... 62302 files and directories currently installed.)
Preparing to unpack .../wireguard-tools_1.0.20200513-1-20.04.2_amd64.deb ...
Unpacking wireguard-tools (1.0.20200513-1-20.04.2) ...
Selecting previously unselected package wireguard.
Preparing to unpack .../wireguard_1.0.20200513-1-20.04.2_all.deb ...
Unpacking wireguard (1.0.20200513-1-20.04.2) ...
Setting up wireguard-tools (1.0.20200513-1-20.04.2) ...
wg-quick.target is a disabled or a static unit, not starting it.
Setting up wireguard (1.0.20200513-1-20.04.2) ...
Processing triggers for man-db (2.9.1-1) ...
agnervillafabrega@wg-server:~$

```

Generar claves del servidor

```

root@wg-server:/home/agnervillafabrega# cd /etc/wireguard
root@wg-server:/etc/wireguard# umask 077
root@wg-server:/etc/wireguard# ls
root@wg-server:/etc/wireguard# wg genkey | sudo tee server_private.key
SHaxS5zYYiqJZeKu+yatyeZBTrgCwBg/Q57yt6pdU4=
root@wg-server:/etc/wireguard# sudo cat server_private.key | wg pubkey | sudo tee server_public.key
OngJgxrlBXdmWLmk2ZujwwBI0TruYemegsveoB6VDk0=
root@wg-server:/etc/wireguard# SERVER_PRIVATE_KEY=$(sudo cat server_private.key)
root@wg-server:/etc/wireguard# SERVER_PUBLIC_KEY=$(sudo cat server_public.key)

```

Clave privada del servidor: SHaxS5zYYiqJZeKu+yatyeZBTrgCwBg/
Q57yt6pdU4=

Clave pública del servidor: OngJgxrlBXdmWLmk2ZujwwBI0TruYeme
gsveoB6VDk0=

Archivo de configuracion:

```

root@wg-server:/etc/wireguard# nano /etc/wireguard/wg0.conf
root@wg-server:/etc/wireguard# cat /etc/wireguard/wg0.conf
[Interface]
# Dirección IP del servidor en la VPN
Address = 10.100.0.1/24
# Puerto de escucha
ListenPort = 51820
# Clave privada del servidor
PrivateKey = SHaxS5zYYiqJZeKu+yatyeZBTrgCwBg/Q57yt6pdU4=
# Comandos post-up y post-down para NAT (opcional)
PostUp = iptables -A FORWARD -i wg0 -j ACCEPT; iptables -t nat -A POSTROUTING -o ens4 -j MASQUERADE
PostDown = iptables -D FORWARD -i wg0 -j ACCEPT; iptables -t nat -D POSTROUTING -o ens4 -j MASQUERADE

[Peer]
# Cliente
PublicKey = +aJT4Fj/b27U1ZXoBuAwEoli2i9QNo0cySYjG3+OyWE=
# IP permitida del cliente
AllowedIPs = 10.100.0.2/32
root@wg-server:/etc/wireguard#

```

Habilitamos el IP forwarding

```

root@wg-server:/etc/wireguard# sudo sysctl -w net.ipv4.ip_forward=1
net.ipv4.ip_forward = 1

```

Habilitamos e iniciamos wireguard

```
root@wg-server:/etc/wireguard# sudo systemctl enable wg-quick@wg0
Created symlink /etc/systemd/system/multi-user.target.wants/wg-quick@wg0.service → /lib/systemd/system/wg-quick@.service.
root@wg-server:/etc/wireguard# sudo systemctl start wg-quick@wg0
root@wg-server:/etc/wireguard# sudo wg show
interface: wg0
  public key: 0ngJgxrlEXdmWlmk2ZujwwB10TruYemegsveoB6VDk0=
  private key: (hidden)
  listening port: 51820

peer: +a0T4Fj/b27U1ZXoBuAwEol12i9QNo0cySYjG3+OywE=
  allowed ips: 10.100.0.2/32
root@wg-server:/etc/wireguard#
```

▼ Configuración de la Cuenta Tercero

Creamos le proyecto

```
agnerfabrega@cloudshell:~ (lexical-acolyte-460722-t6)$ gcloud projects create tercero-proyecto-vpn --name="Tercero Proyecto VPN"
Create in progress for [https://cloudresourcemanager.googleapis.com/v1/projects/tercero-proyecto-vpn].
Waiting for [operations/create_project.global.5186214397543091813] to finish...done.
Enabling service [cloudapis.googleapis.com] on project [tercero-proyecto-vpn]...
Operation "operations/acat.p2-1015109195777-0394562a-cb81-42eb-9776-24176fddb90c" finished successfully.
agnerfabrega@cloudshell:~ (lexical-acolyte-460722-t6)$ gcloud config set project tercero-proyecto-vpn
Updated property [core/project].
agnerfabrega@cloudshell:~ (tercero-proyecto-vpn)$
```

Cambiamos la facturación (Paso anterior en el cli ahora en la consola de gcp):

Google Cloud

Facturación / Proyectos

Facturación

Mis cuentas de facturación Mis proyectos

Filtro Ingresar el nombre o el valor de la propiedad

Nombre	ID	Cuenta de facturación ↑	ID de cuenta de facturación	Acciones ?
Gemini API	gen-lang-client-0851952701	La facturación está inhabilitada	—	⋮
My First Project	lexical-	La facturación está	—	⋮
Tercero Proyecto VPN				⋮

Establece la cuenta de facturación del proyecto "Tercero Proyecto VPN"

Cuenta de facturación *

Mi cuenta de facturación

Todos los cargos de este proyecto se realizarán a la cuenta que selecciones aquí.

Cancelar Establecer cuenta

Activamos el servicio de computo


```

agnerfabrega@cloudshell:~ (tercero-proyecto-vpn)$ gcloud services enable compute.googleapis.com
Operation "operations/acf.p2-1015109195777-36e715ea-f0c0-4445-8262-0708cd04225f" finished successfully.
agnerfabrega@cloudshell:~ (tercero-proyecto-vpn)$

```

VPC

```

agnerfabrega@cloudshell:~ (tercero-proyecto-vpn)$ gcloud compute networks create vpc-tercero \
  --subnet-mode=custom \
  --bgp-routing-mode=regional
Created [https://www.googleapis.com/compute/v1/projects/tercero-proyecto-vpn/global/networks/vpc-tercero].
NAME: vpc-tercero
SUBNET MODE: CUSTOM
BGP ROUTING MODE: REGIONAL
IPV4 RANGE:
GATEWAY_IPV4:
INTERNAL_IPV6_RANGE:

Instances on this network will not be reachable until firewall rules
are created. As an example, you can allow all internal traffic between
instances as well as SSH, RDP, and ICMP by running:

$ gcloud compute firewall-rules create <FIREWALL_NAME> --network vpc-tercero --allow tcp,udp,icmp --source-ranges <I
P_RANGE>
$ gcloud compute firewall-rules create <FIREWALL_NAME> --network vpc-tercero --allow tcp:22,tcp:3389,icmp

agnerfabrega@cloudshell:~ (tercero-proyecto-vpn)$

```

SUBNET

```

agnerfabrega@cloudshell:~ (tercero-proyecto-vpn)$ gcloud compute networks subnets create subnet-tercero \
  --network=vpc-tercero \
  --region=us-central1 \
  --range=10.0.2.0/24
Created [https://www.googleapis.com/compute/v1/projects/tercero-proyecto-vpn/regions/us-central1/subnetworks/subnet-tercero].
NAME: subnet-tercero
REGION: us-central1
NETWORK: vpc-tercero
RANGE: 10.0.2.0/24
STACK TYPE: IPV4_ONLY
IPV6 ACCESS TYPE:
INTERNAL_IPV6_PREFIX:
EXTERNAL_IPV6_PREFIX:
agnerfabrega@cloudshell:~ (tercero-proyecto-vpn)$

```

FIREWALL (WG & SSH)

```

EXTERNAL_IPV6_PREFIX:
agnerfabrega@cloudshell:~ (tercero-proyecto-vpn)$ gcloud compute firewall-rules create allow-wireguard-tercero \
  --network=vpc-tercero \
  --allow=udp:51820 \
  --source-ranges=0.0.0.0/0
Creating firewall...working..Created [https://www.googleapis.com/compute/v1/projects/tercero-proyecto-vpn/global/firewalls/allow-wireguard-tercero].
Creating firewall...done.
NAME: allow-wireguard-tercero
NETWORK: vpc-tercero
DIRECTION: INGRESS
PRIORITY: 1000
ALLOW: udp:51820
DENY:
DISABLED: False
agnerfabrega@cloudshell:~ (tercero-proyecto-vpn)$

```

```

agnerfabrega@cloudshell:~ (tercero-proyecto-vpn)$ gcloud compute firewall-rules create allow-ssh-tercero \
  --network=vpc-tercero \
  --allow=tcp:22 \
  --source-ranges=0.0.0.0/0
Creating firewall...working..Created [https://www.googleapis.com/compute/v1/projects/tercero-proyecto-vpn/global/firewalls/allow-ssh-tercero].
Creating firewall...done.
NAME: allow-ssh-tercero
NETWORK: vpc-tercero
DIRECTION: INGRESS
PRIORITY: 1000
ALLOW: tcp:22
DENY:
DISABLED: False
agnerfabrega@cloudshell:~ (tercero-proyecto-vpn)$

```

Creacion de instancia

```
agnerfabrega@cloudshell:~ (tercero-proyecto-vpn)$ gcloud compute instances create wg-client \
--zone=us-central1-a \
--machine-type=e2-micro \
--network-interface=network=vpn-tercero,subnet=subnet-tercero \
--image-family=ubuntu-2004-lts \
--image-project=ubuntu-os-cloud \
--boot-disk-size=10GB \
--tags=wireguard-client
WARNING: You have selected a disk size of under [200GB]. This may result in poor I/O performance. For more informati
on, see: https://developers.google.com/compute/docs/disks#performance.
Created [https://www.googleapis.com/compute/v1/projects/tercero-proyecto-vpn/zones/us-central1-a/instances/wg-client
].
NAME: wg-client
ZONE: us-central1-a
MACHINE TYPE: e2-micro
PREEMPTIBLE:
INTERNAL IP: 10.0.2.2
EXTERNAL IP: 104.155.152.152
STATUS: RUNNING
agnerfabrega@cloudshell:~ (tercero-proyecto-vpn)$
```

Conexion e instalacion de wireGuard

Conexion a la instancia

```
agnerfabrega@cloudshell:~ (tercero-proyecto-vpn)$ gcloud compute ssh wg-client --zone=us-central1-a
WARNING: The private SSH key file for gcloud does not exist.
WARNING: The public SSH key file for gcloud does not exist.
WARNING: You do not have an SSH key for gcloud.
WARNING: SSH keygen will be executed to generate a key.
This tool needs to create the directory [/home/agnerfabrega/.ssh] before being
able to generate SSH keys.

Do you want to continue (Y/n)? Y

Generating public/private rsa key pair.
Enter passphrase (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /home/agnerfabrega/.ssh/google_compute_engine
Your public key has been saved in /home/agnerfabrega/.ssh/google_compute_engine.pub
The key fingerprint is:
SHA256:CL/M5ti4j6rSXPNLzVePClXI2ifwLqchXSTIWqGX5h8 agnerfabrega@cs-222983499408-default
The key's randomart image is:
+---[RSA 3072]-----+
|      ..o      |
|      .+o...   |
|      .o= oo.   |
|      o.= =..   |
|      o S.E.o   |
|      + B.+ = o |
|      + B *.= . |
| o . * . *..   |
|oo+.o.+....   |
+---[SHA256]-----+
Updating project ssh metadata...working.Updated [https://www.googleapis.com/compute/v1/projects/tercero-proyecto-vpn
].
Updating project ssh metadata...done.
Waiting for SSH key to propagate.
Warning: Permanently added 'compute.6027470060168082669' (ED25519) to the list of known hosts.
Welcome to Ubuntu 20.04.6 LTS (GNU/Linux 5.15.0-1081-gcp x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Sun May 25 17:05:31 UTC 2025

System load:  0.0          Processes:      102
Usage of /:   21.4% of 9.51GB    Users logged in:  0
Memory usage: 21%          IPv4 address for ens4: 10.0.2.2
Swap usage:   0%

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status
```

Instalacion de Wireguard

```

agnerfabrega@wg-client:~$ sudo apt install -y wireguard
Reading package lists... Done
Building dependency tree
Reading state information... Done
The following additional packages will be installed:
  wireguard-tools
Suggested packages:
  openresolv | resolvconf
The following NEW packages will be installed:
  wireguard wireguard-tools
0 upgraded, 2 newly installed, 0 to remove and 7 not upgraded.
Need to get 86.6 kB of archives.
After this operation, 344 kB of additional disk space will be used.
Get:1 http://us-central1.gce.archive.ubuntu.com/ubuntu focal-updates/universe amd64 wireguard-tools amd64 1.0.20200513-1-20.04.2 [83.3 kB]
Get:2 http://us-central1.gce.archive.ubuntu.com/ubuntu focal-updates/universe amd64 wireguard all 1.0.20200513-1-20.04.2 [3264 B]
Fetched 86.6 kB in 0s (855 kB/s)
Selecting previously unselected package wireguard-tools.
(Reading database ... 62302 files and directories currently installed.)
Preparing to unpack .../wireguard-tools 1.0.20200513-1-20.04.2_amd64.deb ...
Unpacking wireguard-tools (1.0.20200513-1-20.04.2) ...
Selecting previously unselected package wireguard.
Preparing to unpack .../wireguard 1.0.20200513-1-20.04.2_all.deb ...
Unpacking wireguard (1.0.20200513-1-20.04.2) ...
Setting up wireguard-tools (1.0.20200513-1-20.04.2) ...
wg-quick.target is a disabled or a static unit, not starting it.
Setting up wireguard (1.0.20200513-1-20.04.2) ...
Processing triggers for man-db (2.9.1-1) ...
agnerfabrega@wg-client:~$

```

Generar claves del servidor

```

agnerfabrega@wg-client:~$ sudo su
root@wg-client:/home/agnerfabrega# cd /etc/wireguard
root@wg-client:/etc/wireguard# umask 077
root@wg-client:/etc/wireguard# wg genkey | sudo tee client_private.key
yGkpeuF2MYJRF3lmgR01dyyfe9hs/2X0EuC3d1LuP3w=
root@wg-client:/etc/wireguard# sudo cat client_private.key | wg pubkey | sudo tee client_public.key
+aJT4Fj/b27U12XoBuAwEoli2i9QNo0cySYjG3+OywE=

```

Clave privada del cliente: yGkpeuF2MYJRF3lmgR01dyyfe9hs/2X0EuC3d1LuP3w=

Clave pública del cliente: +aJT4Fj/b27U12XoBuAwEoli2i9QNo0cySYjG3+OywE=

Archivo de configuracion:

```

agnerfabrega@wg-client:~$ sudo su
root@wg-client:/home/agnerfabrega# nano /etc/wireguard/wg0.conf
root@wg-client:/home/agnerfabrega# cat /etc/wireguard/wg0.conf
[Interface]
# Dirección IP del cliente en la VPN
Address = 10.100.0.2/24
# Clave privada del cliente (generada en el Paso 2.2)
PrivateKey = yGkpeuF2MYJRF3lmgR01dyffe9hs/2X0EuC3d1LuP3w=
# DNS (opcional)
DNS = 8.8.8.8

[Peer]
# Información del servidor (cuenta propia)
# Clave pública del servidor (generada en el Paso 2.1)
PublicKey = OngJgxrlBXdmWLmk2ZujwwB10TruYemegsveoB6VDk0=
# IP pública del servidor:Puerto (obtenida en 4.1)
Endpoint = 35.231.249.130:51820
# Rango de IPs que se enrutarán a través del túnel
AllowedIPs = 10.100.0.0/24
# Mantener la conexión activa (envía paquete cada 25 segundos)
PersistentKeepalive = 25
root@wg-client:/home/agnerfabrega#

```

```

root@wg-client:/home/agnerfabrega# wg show
interface: wg0
  public key: +aJT4Fj/b27U1ZXoBuAwEoli2i9QNo0cySYjG3+OywE=
  private key: (hidden)
  listening port: 47889

peer: OngJgxrlBXdmWLmk2ZujwwB10TruYemegsveoB6VDk0=
  endpoint: 34.68.174.17:51820
  allowed ips: 10.100.0.0/24
  transfer: 0 B received, 296 B sent
  persistent keepalive: every 25 seconds
root@wg-client:/home/agnerfabrega#

```

▼ Errores a los que me enfrente

Error al levantar el servicio de WireGuard en el la instanacia del cliente, en resumidas cuentas el servicio falla porque está intentando usar el comando `resolvconf`, que **no está instalado**.

```

root@wg-client:/home/agnerfabrega# ls -la /etc/wireguard/wg0.conf
-rw-r--r-- 1 root root 637 May 25 20:16 /etc/wireguard/wg0.conf
root@wg-client:/home/agnerfabrega# sudo systemctl enable wg-quick@wg0
Created symlink /etc/systemd/system/multi-user.target.wants/wg-quick@wg0.service → /lib/systemd/system/wg-quick@.service.
root@wg-client:/home/agnerfabrega# sudo systemctl start wg-quick@wg0
Job for wg-quick@wg0.service failed because the control process exited with error code.
See "systemctl status wg-quick@wg0.service" and "journalctl -xe" for details.
root@wg-client:/home/agnerfabrega# sudo systemctl status wg-quick@wg0
● wg-quick@wg0.service - WireGuard via wg-quick(8) for wg0
   Loaded: loaded (/lib/systemd/system/wg-quick@.service; enabled; vendor preset: enabled)
   Active: failed (Result: exit-code) since Sun 2025-05-25 20:19:47 UTC; 47s ago
     Docs: man:wg-quick(8)
           man:wg(8)
           https://www.wireguard.com/
           https://www.wireguard.com/quickstart/
           https://git.zx2c4.com/wireguard-tools/about/src/man/wg-quick.8
           https://git.zx2c4.com/wireguard-tools/about/src/man/wg.8
   Process: 14439 ExecStart=/usr/bin/wg-quick up wg0 (code=exited, status=127)
   Main PID: 14439 (code=exited, status=127)

May 25 20:19:47 wg-client wg-quick[14439]: [#] ip link add wg0 type wireguard
May 25 20:19:47 wg-client wg-quick[14439]: [#] wg setconf wg0 /dev/fd/63
May 25 20:19:47 wg-client wg-quick[14439]: [#] ip -4 address add 10.100.0.2/24 dev wg0
May 25 20:19:47 wg-client wg-quick[14439]: [#] ip link set mtu 1380 up dev wg0
May 25 20:19:47 wg-client wg-quick[14475]: [#] resolvconf -a wg0 -m 0 -x
May 25 20:19:47 wg-client wg-quick[14477]: /usr/bin/wg-quick: line 32: resolvconf: command not found
May 25 20:19:47 wg-client wg-quick[14439]: [#] ip link delete dev wg0
May 25 20:19:47 wg-client systemd[1]: wg-quick@wg0.service: Main process exited, code=exited, status=127/n/a
May 25 20:19:47 wg-client systemd[1]: wg-quick@wg0.service: Failed with result 'exit-code'.
May 25 20:19:47 wg-client systemd[1]: Failed to start WireGuard via wg-quick(8) for wg0.

```

Este comando es usado por `wg-quick` cuando nuestro archivo `wg0.conf` tiene una asignación de DNS como:

```
DNS = 1.1.1.1
```

La solución fue sencilla solo necesitaba instalar `resolvconf`

```

sudo apt update
sudo apt install resolvconf

#Después reiniciamos el servicio:
sudo systemctl restart wg-quick@wg0

```

Servicio funcionando correctamente

```

root@wg-client:/home/agnerfabrega# systemctl enable wg-quick@wg0
root@wg-client:/home/agnerfabrega# systemctl start wg-quick@wg0
root@wg-client:/home/agnerfabrega# systemctl status wg-quick@wg0
● wg-quick@wg0.service - WireGuard via wg-quick(8) for wg0
   Loaded: loaded (/lib/systemd/system/wg-quick@.service; enabled; vendor preset: enabled)
   Active: active (exited) since Sun 2025-05-25 20:21:57 UTC; 41s ago
     Docs: man:wg-quick(8)
           man:wg(8)
           https://www.wireguard.com/
           https://www.wireguard.com/quickstart/
           https://git.zx2c4.com/wireguard-tools/about/src/man/wg-quick.8
           https://git.zx2c4.com/wireguard-tools/about/src/man/wg.8
   Main PID: 14900 (code=exited, status=0/SUCCESS)
    Tasks: 0 (limit: 1134)
   Memory: 0B
   CGroup: /system.slice/system-wg\x2dquick.slice/wg-quick@wg0.service

May 25 20:21:57 wg-client systemd[1]: Starting WireGuard via wg-quick(8) for wg0...
May 25 20:21:57 wg-client wg-quick[14900]: [#] ip link add wg0 type wireguard
May 25 20:21:57 wg-client wg-quick[14900]: [#] wg setconf wg0 /dev/fd/63
May 25 20:21:57 wg-client wg-quick[14900]: [#] ip -4 address add 10.100.0.2/24 dev wg0
May 25 20:21:57 wg-client wg-quick[14900]: [#] ip link set mtu 1380 up dev wg0
May 25 20:21:57 wg-client wg-quick[14925]: [#] resolvconf -a tun.wg0 -m 0 -x
May 25 20:21:57 wg-client systemd[1]: Finished WireGuard via wg-quick(8) for wg0.
root@wg-client:/home/agnerfabrega#

```

▼ Validacion y prueba a la VPN

Validacion del tunel:

Servidor principal:

```

(mi-proyecto-vpn-460916) x + ▾
Abrir editor
agnervillafabrega@wg-server:~$ sudo wg show
interface: wg0
  public key: 0mgJgxrlBXdmWLmk2ZujwwB10TruYemegsveoB6VDk0=
  private key: (hidden)
  listening port: 51820

peer: +aJT4Fj/b27ULZXoBuAwEol12i9QNo0cySYjG3+OywE=
  endpoint: 104.155.152.152:59293
  allowed ips: 10.100.0.2/32
  latest handshake: 55 seconds ago
  transfer: 1.07 KiB received, 696 B sent
agnervillafabrega@wg-server:~$ sudo journalctl -u wg-quick@wg0 -f
-- Logs begin at Sun 2025-05-25 16:39:51 UTC. --
May 25 20:51:44 wg-server wg-quick[14924]: [#] iptables -D FORWARD -i wg0 -j ACCEPT; iptables -t nat -D POSTROUTING
-o ens4 -j MASQUERADE
May 25 20:51:44 wg-server systemd[1]: wg-quick@wg0.service: Succeeded.
May 25 20:51:44 wg-server systemd[1]: Stopped WireGuard via wg-quick(8) for wg0.
May 25 20:51:44 wg-server systemd[1]: Starting WireGuard via wg-quick(8) for wg0...
May 25 20:51:44 wg-server wg-quick[14945]: [#] ip link add wg0 type wireguard
May 25 20:51:44 wg-server wg-quick[14945]: [#] wg setconf wg0 /dev/fd/63
May 25 20:51:44 wg-server wg-quick[14945]: [#] ip -4 address add 10.100.0.1/24 dev wg0
May 25 20:51:44 wg-server wg-quick[14945]: [#] ip link set mtu 1380 up dev wg0
May 25 20:51:44 wg-server wg-quick[14945]: [#] iptables -A FORWARD -i wg0 -j ACCEPT; iptables -t nat -A POSTROUTING
-o ens4 -j MASQUERADE
May 25 20:51:44 wg-server systemd[1]: Finished WireGuard via wg-quick(8) for wg0.

```

Servidor del cliente:

```
(tercero-proyecto-vpn) X + v
Abrir editor

root@wg-client:/home/agnerfabrega# ping -c 4 10.100.0.1
PING 10.100.0.1 (10.100.0.1) 56(84) bytes of data.
64 bytes from 10.100.0.1: icmp_seq=1 ttl=64 time=3.81 ms
64 bytes from 10.100.0.1: icmp_seq=2 ttl=64 time=0.918 ms
64 bytes from 10.100.0.1: icmp_seq=3 ttl=64 time=1.17 ms
64 bytes from 10.100.0.1: icmp_seq=4 ttl=64 time=1.27 ms

--- 10.100.0.1 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3005ms
rtt min/avg/max/mdev = 0.918/1.791/3.813/1.174 ms
root@wg-client:/home/agnerfabrega# ip addr show wg0
9: wg0: <POINTOPOINT,NOARP,UP,LOWER_UP> mtu 1380 qdisc noqueue state UNKNOWN group default qlen 1000
    link/none
        inet 10.100.0.2/24 scope global wg0
            valid_lft forever preferred_lft forever
root@wg-client:/home/agnerfabrega# sudo wg show
sudo: unable to resolve host wg-client: Name or service not known
interface: wg0
    public key: +aJT4Fj/b27U1ZXoBuAwEoli2i9QN0cYsYjG3+OywE=
    private key: (hidden)
    listening port: 59293

peer: OngJgxrlBXdmWlMk2ZujwwB10TruYemegsveoB6VDk0=
    endpoint: 34.68.174.176:51820
    allowed ips: 10.100.0.0/24
    latest handshake: 1 minute, 43 seconds ago
    transfer: 604 B received, 820 B sent
    persistent keepalive: every 25 seconds
```

Verificacion tunel establecido (Generamos trafico de red por la VPN)

```
(mi-proyecto-vpn-460916) X + v
Abrir editor

agnergillafabrega@wg-server:~$ sudo tcpdump -i wg0 -n
tcpdump: verbose output suppressed, use -v or -vv for full protocol decode
listening on wg0, link-type RAW (Raw IP), capture size 262144 bytes
23:05:09.684872 IP 10.100.0.2 > 10.100.0.1: ICMP echo request, id 13, seq 1, length 64
23:05:09.684933 IP 10.100.0.1 > 10.100.0.2: ICMP echo reply, id 13, seq 1, length 64
23:05:10.684301 IP 10.100.0.2 > 10.100.0.1: ICMP echo request, id 13, seq 2, length 64
23:05:10.684346 IP 10.100.0.1 > 10.100.0.2: ICMP echo reply, id 13, seq 2, length 64
23:05:11.685360 IP 10.100.0.2 > 10.100.0.1: ICMP echo request, id 13, seq 3, length 64
23:05:11.685391 IP 10.100.0.1 > 10.100.0.2: ICMP echo reply, id 13, seq 3, length 64
23:05:12.687050 IP 10.100.0.2 > 10.100.0.1: ICMP echo request, id 13, seq 4, length 64
23:05:12.687091 IP 10.100.0.1 > 10.100.0.2: ICMP echo reply, id 13, seq 4, length 64
23:05:13.688423 IP 10.100.0.2 > 10.100.0.1: ICMP echo request, id 13, seq 5, length 64
23:05:13.688491 IP 10.100.0.1 > 10.100.0.2: ICMP echo reply, id 13, seq 5, length 64
23:05:14.689822 IP 10.100.0.2 > 10.100.0.1: ICMP echo request, id 13, seq 6, length 64
23:05:14.689856 IP 10.100.0.1 > 10.100.0.2: ICMP echo reply, id 13, seq 6, length 64
23:05:15.691368 IP 10.100.0.2 > 10.100.0.1: ICMP echo request, id 13, seq 7, length 64
23:05:15.691434 IP 10.100.0.1 > 10.100.0.2: ICMP echo reply, id 13, seq 7, length 64
23:05:16.693562 IP 10.100.0.2 > 10.100.0.1: ICMP echo request, id 13, seq 8, length 64
23:05:16.693631 IP 10.100.0.1 > 10.100.0.2: ICMP echo reply, id 13, seq 8, length 64
23:05:17.695256 IP 10.100.0.2 > 10.100.0.1: ICMP echo request, id 13, seq 9, length 64
23:05:17.695326 IP 10.100.0.1 > 10.100.0.2: ICMP echo reply, id 13, seq 9, length 64
23:05:18.697262 IP 10.100.0.2 > 10.100.0.1: ICMP echo request, id 13, seq 10, length 64
23:05:18.697325 IP 10.100.0.1 > 10.100.0.2: ICMP echo reply, id 13, seq 10, length 64
23:05:19.698426 IP 10.100.0.2 > 10.100.0.1: ICMP echo request, id 13, seq 11, length 64
23:05:19.698674 IP 10.100.0.1 > 10.100.0.2: ICMP echo reply, id 13, seq 11, length 64
23:05:20.700296 IP 10.100.0.2 > 10.100.0.1: ICMP echo request, id 13, seq 12, length 64
23:05:20.700357 IP 10.100.0.1 > 10.100.0.2: ICMP echo reply, id 13, seq 12, length 64
23:05:21.701675 IP 10.100.0.2 > 10.100.0.1: ICMP echo request, id 13, seq 13, length 64
23:05:21.701719 IP 10.100.0.1 > 10.100.0.2: ICMP echo reply, id 13, seq 13, length 64
23:05:22.703237 IP 10.100.0.2 > 10.100.0.1: ICMP echo request, id 13, seq 14, length 64
23:05:22.703296 IP 10.100.0.1 > 10.100.0.2: ICMP echo reply, id 13, seq 14, length 64
23:05:23.704587 IP 10.100.0.2 > 10.100.0.1: ICMP echo request, id 13, seq 15, length 64
23:05:23.704628 IP 10.100.0.1 > 10.100.0.2: ICMP echo reply, id 13, seq 15, length 64
23:05:24.706154 IP 10.100.0.2 > 10.100.0.1: ICMP echo request, id 13, seq 16, length 64
23:05:24.706208 IP 10.100.0.1 > 10.100.0.2: ICMP echo reply, id 13, seq 16, length 64
23:05:25.707553 IP 10.100.0.2 > 10.100.0.1: ICMP echo request, id 13, seq 17, length 64
23:05:25.707611 IP 10.100.0.1 > 10.100.0.2: ICMP echo reply, id 13, seq 17, length 64
23:05:26.709436 IP 10.100.0.2 > 10.100.0.1: ICMP echo request, id 13, seq 18, length 64
23:05:26.709500 IP 10.100.0.1 > 10.100.0.2: ICMP echo reply, id 13, seq 18, length 64
23:05:27.711347 IP 10.100.0.2 > 10.100.0.1: ICMP echo request, id 13, seq 19, length 64
23:05:27.711412 IP 10.100.0.1 > 10.100.0.2: ICMP echo reply, id 13, seq 19, length 64
23:05:28.713361 IP 10.100.0.2 > 10.100.0.1: ICMP echo request, id 13, seq 20, length 64
23:05:28.713416 IP 10.100.0.1 > 10.100.0.2: ICMP echo reply, id 13, seq 20, length 64
[]

(tercero-proyecto-vpn) X + v
Abrir editor

root@wg-client:/home/agnerfabrega# ping 10.100.0.1
PING 10.100.0.1 (10.100.0.1) 56(84) bytes of data.
64 bytes from 10.100.0.1: icmp_seq=1 ttl=64 time=1.83 ms
64 bytes from 10.100.0.1: icmp_seq=2 ttl=64 time=1.06 ms
64 bytes from 10.100.0.1: icmp_seq=3 ttl=64 time=1.10 ms
64 bytes from 10.100.0.1: icmp_seq=4 ttl=64 time=1.15 ms
64 bytes from 10.100.0.1: icmp_seq=5 ttl=64 time=1.03 ms
64 bytes from 10.100.0.1: icmp_seq=6 ttl=64 time=1.27 ms
64 bytes from 10.100.0.1: icmp_seq=7 ttl=64 time=1.42 ms
64 bytes from 10.100.0.1: icmp_seq=8 ttl=64 time=1.68 ms
64 bytes from 10.100.0.1: icmp_seq=9 ttl=64 time=1.74 ms
64 bytes from 10.100.0.1: icmp_seq=10 ttl=64 time=1.14 ms
64 bytes from 10.100.0.1: icmp_seq=11 ttl=64 time=1.36 ms
64 bytes from 10.100.0.1: icmp_seq=12 ttl=64 time=1.25 ms
64 bytes from 10.100.0.1: icmp_seq=13 ttl=64 time=1.19 ms
64 bytes from 10.100.0.1: icmp_seq=14 ttl=64 time=1.21 ms
64 bytes from 10.100.0.1: icmp_seq=15 ttl=64 time=1.13 ms
64 bytes from 10.100.0.1: icmp_seq=16 ttl=64 time=1.12 ms
64 bytes from 10.100.0.1: icmp_seq=17 ttl=64 time=1.32 ms
64 bytes from 10.100.0.1: icmp_seq=18 ttl=64 time=1.65 ms
64 bytes from 10.100.0.1: icmp_seq=19 ttl=64 time=1.66 ms
64 bytes from 10.100.0.1: icmp_seq=20 ttl=64 time=1.26 ms
[]
```

Prueba desde el servidor principal hacia nuestro cliente:


```
6 packets dropped by kernel
agnervillafabrega@wg-server:~$ ping -c 10 10.100.0.2
PING 10.100.0.2 (10.100.0.2) 56(84) bytes of data.
64 bytes from 10.100.0.2: icmp_seq=1 ttl=64 time=2.32 ms
64 bytes from 10.100.0.2: icmp_seq=2 ttl=64 time=1.23 ms
64 bytes from 10.100.0.2: icmp_seq=3 ttl=64 time=1.19 ms
64 bytes from 10.100.0.2: icmp_seq=4 ttl=64 time=1.28 ms
64 bytes from 10.100.0.2: icmp_seq=5 ttl=64 time=1.28 ms
64 bytes from 10.100.0.2: icmp_seq=6 ttl=64 time=1.56 ms
^C
--- 10.100.0.2 ping statistics ---
6 packets transmitted, 6 received, 0% packet loss, time 5008ms
rtt min/avg/max/mdev = 1.188/1.476/2.318/0.394 ms
agnervillafabrega@wg-server:~$
```

Prueba con transferencia de archivos

```
agnervillafabrega@wg-server:~$ echo "Prueba de transferencia VPS" > /tmp/prueba.txt
agnervillafabrega@wg-server:~$ cd /tmp && python3 -m http.server 8000
Serving HTTP on 0.0.0.0 port 8000 (http://0.0.0.0:8000/) ...
10.100.0.2 - - [25/May/2023 23:08:11] "GET /prueba.txt HTTP/1.1" 200 -

root@wg-client:/home/agnervillafabrega# wget http://10.100.0.1:8000/prueba.txt -O /tmp/prueba_recibida.txt
--2023-05-25 23:08:11-- http://10.100.0.1:8000/prueba.txt
Connecting to 10.100.0.1:8000... connected.
HTTP request sent, awaiting response... 200 OK
Length: 28 [text/plain]
Saving to: '/tmp/prueba_recibida.txt'

/tmp/prueba_recibida.txt[=====] 28 ---KB/s in 0s

2023-05-25 23:08:11 (4.15 MB/s) - '/tmp/prueba_recibida.txt' saved [28/28]

root@wg-client:/home/agnervillafabrega# cat /tmp/prueba_recibida.txt
Prueba de transferencia VPS
root@wg-client:/home/agnervillafabrega#
```

Prueba de rendimiento

```
agnervillafabrega@wg-server:~$ iperf3 -s
Server listening on 5201

Accepted connection from 10.100.0.2, port 53778
[ ID] Interval      Transfer     Bitrate
[ 5] 0.00-1.00 sec  85.3 MBytes  716 Mbits/sec
[ ID] Interval      Transfer     Bitrate
[ 5] 1.00-2.00 sec  108 MBytes  903 Mbits/sec
[ 5] 2.00-3.00 sec  108 MBytes  904 Mbits/sec
[ 5] 3.00-4.00 sec  108 MBytes  903 Mbits/sec
[ 5] 4.00-5.00 sec  108 MBytes  904 Mbits/sec
[ 5] 5.00-6.00 sec  104 MBytes  873 Mbits/sec
[ 5] 6.00-7.00 sec  101 MBytes  844 Mbits/sec
[ 5] 7.00-8.00 sec  108 MBytes  905 Mbits/sec
[ 5] 8.00-9.00 sec  108 MBytes  904 Mbits/sec
[ 5] 9.00-10.00 sec  108 MBytes  904 Mbits/sec
[ 5] 10.00-11.00 sec  108 MBytes  905 Mbits/sec
[ 5] 11.00-12.00 sec  108 MBytes  904 Mbits/sec
[ 5] 12.00-13.00 sec  108 MBytes  904 Mbits/sec
[ 5] 13.00-14.00 sec  108 MBytes  904 Mbits/sec
[ 5] 14.00-15.00 sec  108 MBytes  904 Mbits/sec
[ 5] 15.00-16.00 sec  108 MBytes  904 Mbits/sec
[ 5] 16.00-17.00 sec  107 MBytes  890 Mbits/sec
[ 5] 17.00-18.00 sec  107 MBytes  901 Mbits/sec
[ 5] 18.00-19.00 sec  108 MBytes  888 Mbits/sec
[ 5] 19.00-20.00 sec  104 MBytes  870 Mbits/sec
[ 5] 20.00-21.00 sec  107 MBytes  901 Mbits/sec
[ 5] 21.00-22.00 sec  108 MBytes  910 Mbits/sec
[ 5] 22.00-23.00 sec  107 MBytes  899 Mbits/sec
[ 5] 23.00-24.00 sec  108 MBytes  904 Mbits/sec
[ 5] 24.00-25.00 sec  23.2 MBytes  196 Mbits/sec
[ 5] 25.00-26.00 sec  27.2 MBytes  228 Mbits/sec
[ 5] 26.00-27.00 sec  22.5 MBytes  189 Mbits/sec
[ 5] 27.00-28.00 sec  20.1 MBytes  169 Mbits/sec
[ 5] 28.00-29.00 sec  20.3 MBytes  171 Mbits/sec
[ 5] 29.00-30.00 sec  21.1 MBytes  178 Mbits/sec
[ 5] 30.00-30.11 sec  1.54 MBytes  2.05 Gbits/sec

[ ID] Interval      Transfer     Bitrate
[ 5] 0.00-30.11 sec  2.62 GBytes  746 Mbits/sec

Server listening on 5201

root@wg-client:/home/agnervillafabrega# iperf3 -c 10.100.0.1 -t 30
Connecting to host 10.100.0.1, port 5201
[ ID] Interval      Transfer     Bitrate      Retr      sender
[ 5] 0.00-1.00 sec  81.4 MBytes  666 Mbits/sec  0         454 Kbytes
[ 5] 1.00-2.00 sec  108 MBytes  905 Mbits/sec  0         774 Kbytes
[ 5] 2.00-3.00 sec  108 MBytes  904 Mbits/sec  0         833 Kbytes
[ 5] 3.00-4.00 sec  109 MBytes  911 Mbits/sec  0         978 Kbytes
[ 5] 4.00-5.00 sec  107 MBytes  899 Mbits/sec  0         978 Kbytes
[ 5] 5.00-6.00 sec  104 MBytes  873 Mbits/sec  0         978 Kbytes
[ 5] 6.00-7.00 sec  101 MBytes  849 Mbits/sec  0         978 Kbytes
[ 5] 7.00-8.00 sec  108 MBytes  903 Mbits/sec  0         1.02 Mbytes
[ 5] 8.00-9.00 sec  108 MBytes  907 Mbits/sec  0         1.02 Mbytes
[ 5] 9.00-10.00 sec  108 MBytes  905 Mbits/sec  0         1.02 Mbytes
[ 5] 10.00-11.00 sec  108 MBytes  904 Mbits/sec  0         1.07 Mbytes
[ 5] 11.00-12.00 sec  107 MBytes  901 Mbits/sec  0         1.07 Mbytes
[ 5] 12.00-13.00 sec  109 MBytes  911 Mbits/sec  0         1.07 Mbytes
[ 5] 13.00-14.00 sec  107 MBytes  901 Mbits/sec  0         1.07 Mbytes
[ 5] 14.00-15.00 sec  107 MBytes  901 Mbits/sec  0         1.07 Mbytes
[ 5] 15.00-16.00 sec  109 MBytes  911 Mbits/sec  0         1.07 Mbytes
[ 5] 16.00-17.00 sec  107 MBytes  900 Mbits/sec  0         1.07 Mbytes
[ 5] 17.00-18.00 sec  107 MBytes  901 Mbits/sec  0         1.07 Mbytes
[ 5] 18.00-19.00 sec  108 MBytes  888 Mbits/sec  0         1.07 Mbytes
[ 5] 19.00-20.00 sec  104 MBytes  870 Mbits/sec  0         1.07 Mbytes
[ 5] 20.00-21.00 sec  107 MBytes  901 Mbits/sec  0         1.07 Mbytes
[ 5] 21.00-22.00 sec  108 MBytes  910 Mbits/sec  0         1.07 Mbytes
[ 5] 22.00-23.00 sec  107 MBytes  899 Mbits/sec  0         1.07 Mbytes
[ 5] 23.00-24.00 sec  108 MBytes  904 Mbits/sec  0         1.07 Mbytes
[ 5] 24.00-25.00 sec  23.2 MBytes  196 Mbits/sec  1         3.61 Mbytes
[ 5] 25.00-26.00 sec  26.2 MBytes  220 Mbits/sec  1         3.61 Mbytes
[ 5] 26.00-27.00 sec  22.5 MBytes  189 Mbits/sec  2         3.61 Mbytes
[ 5] 27.00-28.00 sec  21.2 MBytes  178 Mbits/sec  210        1.94 Mbytes
[ 5] 28.00-29.00 sec  20.0 MBytes  168 Mbits/sec  16         1.46 Mbytes
[ 5] 29.00-30.00 sec  21.2 MBytes  178 Mbits/sec  0         1.54 Mbytes

[ ID] Interval      Transfer     Bitrate      Retr      sender
[ 5] 0.00-30.00 sec  2.62 GBytes  746 Mbits/sec  230
[ 5] 0.00-30.11 sec  2.62 GBytes  746 Mbits/sec

iperf3 Done.
root@wg-client:/home/agnervillafabrega#
```

▼ Mejoras y Próximos Pasos

1. Infraestructura como Código (IaC) con Terraform

```
# Ejemplo de módulo Terraform para WireGuard
module "wireguard_vpn" {
  source = "./modules/wireguard"
```



```
# Configuración del servidor
server_project_id = "mi-proyecto-vpn"
server_zone       = "us-central1-a"
server_vpc_cidr   = "10.0.1.0/24"

# Configuración del cliente
client_project_id = "tercero-proyecto-vpn"
client_zone       = "us-central1-a"
client_vpc_cidr   = "10.0.2.0/24"

# Red VPN
vpn_cidr = "10.100.0.0/24"
}
```

Beneficios de usar Terraform:

- Despliegue reproducible y versionado
- Gestión del estado de la infraestructura
- Facilita la creación de múltiples entornos
- Integración con CI/CD

2. Automatización con Ansible

```
# Ejemplo de playbook Ansible para configurar WireGuard
---
- name: Configurar WireGuard VPN
  hosts: wireguard_servers
  become: yes

  tasks:
    - name: Instalar WireGuard
      apt:
        name: wireguard
        state: present
        update_cache: yes

    - name: Generar claves si no existen
```

```
shell: |
  wg genkey | tee /etc/wireguard/{{ inventory_hostname }}.key | wg
  pubkey > /etc/wireguard/{{ inventory_hostname }}.pub
args:
  creates: /etc/wireguard/{{ inventory_hostname }}.key

- name: Configurar WireGuard
  template:
    src: wg0.conf.j2
    dest: /etc/wireguard/wg0.conf
    mode: '0600'
  notify: restart wireguard
```

Ventajas de Ansible:

- Configuración consistente y repetible
- Gestión centralizada de múltiples servidores
- Integración con Vault para secretos
- Idempotencia garantizada

▼ Conclusiones

Se ha implementado exitosamente una conexión VPN segura entre dos servidores ubicados en diferentes cuentas de Google Cloud Platform utilizando WireGuard. La solución cumple con todos los objetivos planteados:

-  **Conexión segura establecida** entre servidores en cuentas GCP separadas
-  **Túnel cifrado funcional** con WireGuard
-  **Comunicación bidireccional** verificada entre las IPs 10.100.0.1 y 10.100.0.2
-  **Gestión segura de claves** con permisos restrictivos